

Data Cleaning & Preprocessing

Task 5 - Data Analyst Internship at Internee.pk
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Executive Summary

This project transforms a realistic portfolio dataset of 900 raw internship applications into 860 structured, analysis-ready records. The automated Python (Pandas) pipeline identifies and handles missing values, duplicate identities, invalid numeric outliers, and inconsistent formatting while preserving a field-level audit trail.

Metric	Result
Raw records	900
Clean records	860
Missing values handled	152
Duplicates removed	40
Outliers treated	32
Standardization fixes	722
Full-dataset quality before	96.7%
Full-dataset quality after	100.0%

Task Requirement Coverage

- Internship application data with applicant demographics and skills
- Missing-value identification and field-specific handling
- Duplicate detection after identity-field normalization
- Outlier / invalid numeric treatment
- Standardization and structuring for efficient analysis
- Python (Pandas) automation with reproducible validation

Cleaning Methodology

1. Data Profiling

Profile row count, schema, missingness, duplicate identity patterns, numeric validity, and inconsistent categories before transformation.

2. Missing Values

CGPA uses a valid median; city uses a province-level mode; gender uses a neutral category; university, skills, phone and LinkedIn use explicit missing categories when a real value cannot be inferred safely.

3. Duplicate Detection

Normalize name, email and phone fields first. Remove repeated identities using email + phone, with name + email as a fallback. Keep the first valid occurrence and retain removed rows in an audit file.

4. Outlier Treatment

Validate age (18-45), CGPA (0-4), experience (0-60 months), and expected stipend (PKR 5,000-100,000). Invalid values are replaced with the median of valid observations for that field.

5. Standardization

Normalize name casing, email casing, phone formatting, city aliases, education labels, department labels, skill delimiters/aliases, LinkedIn URLs, and timestamps.

6. Validation

The project validator reconciles raw/clean counts and confirms duplicate removal, required-field completion, numeric ranges, email normalization, and required deliverable files.

Data Quality Issues

Issue category	Actions
Missing Values	152
Duplicate Records	40
Outliers	32
Formatting / Standardization	722

Dashboard & Decision Support

The browser dashboard uses a distinct data-operations design rather than a basic collection of bars and donuts. It includes a cleaning pipeline, Sankey-style process flow, calendar issue-density heatmap, outlier box plots, treemap-style issue composition, before/after field-quality heatmap, KPI progress, correction audit table, filters, clean-data review, CSV export, report printing, and a dark/light theme toggle.

Key Recommendations

- Apply field validation before application submission.
- Use controlled dropdowns for city, education and department values.
- Normalize identity fields before duplicate checks.
- Run the Pandas cleaning pipeline before each reporting refresh.
- Preserve the cleaning log to maintain traceability and reproducibility.

Ethical & Data Note

The project uses a realistic portfolio dataset for internship training. It does not contain confidential applicant records. Missing values are not fabricated when a defensible value is unavailable; explicit missing categories are used instead. All transformation rules are documented and auditable.

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